

# Exact Finite Dimer Random Sequential Adsorption on Paths and Cycles

Route-A source-local certificate HCS-C291

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## Abstract

We give a finite-size solution of dimer random sequential adsorption on a path and a simple cycle, viewed as a uniformly random greedy edge order. Arbitrary differentiation produces a triangular system for all factorial moments. Explicit first and second solutions give  $\text{Var}(M_n) = e^{-4}n + 2e^{-4} + o(1)$ . Binary gap words give the exact path and cycle supports, with every allowed cardinality attained. A producer-independent bitmask/order oracle, symbolic reconstruction, fresh-path replay, and repaired-hash mutation suite audit the theorem without replacing its all-size proof. The classical stochastic model has no arithmetic prime clock, target bridge, or source-native quantization, so all five Route-A gates fail.

## 1 Model, ownership, and claim boundary

Let  $P_n$  have vertices  $1, \dots, n$  and edges  $e_i = \{i, i+1\}$ , with  $P_0$  empty. Assign iid continuous priorities to the edges and inspect them in increasing order. An edge is accepted exactly when both endpoints are currently unmatched. Equivalently, scan a uniformly random permutation of the labeled edges. Let  $M_n$  be the number accepted and write

$$F_n(z) = \mathbb{E}z^{M_n}.$$

For the simple labeled cycle  $C_n$ ,  $n \geq 3$ , write the corresponding count and PGF as  $K_n$  and  $G_n(z)$ .

The terminal edge set is a jammed *maximal* matching: no further edge can be added. It need not be a *maximum* matching. This distinction is part of every statement below. Continuous priorities merely generate a uniform order; a discrete-priority model needs a separate tie-breaking convention.

One-dimensional blocking goes back at least to Flory's polymer calculation [1]; Evans reviews random and cooperative sequential adsorption [2]. Penrose gives rigorous general jamming-limit context [3], while Dyer and Frieze place randomized greedy matching in its algorithmic lineage [4]. We reconstruct an exact finite theorem and integrity chain. We do not claim invention or literature priority for the model, the limiting density, or random greedy matching.

## 2 The finite theorem

**Theorem 1** (Path/cycle PGFs, moments, support, and boundary effect). *Set  $F_0 = F_1 = 1$ . For every  $n \geq 2$ ,*

$$(n-1)F_n(z) = z \sum_{a+b=n-2} F_a(z)F_b(z). \quad (1)$$

*Thus the formal OGF  $F(x, z) = \sum_{n \geq 0} F_n(z)x^n$  is the unique formal solution with  $[x^0]F = [x^1]F = 1$  of*

$$F_x = \frac{F-1}{x} + zx F^2, \quad xF_x - F + 1 = zx^2 F^2. \quad (2)$$

*Put  $H_r(x) = \partial_z^r F(x, z)|_{z=1}$ . Then  $H_0 = (1-x)^{-1}$ , and for every  $r \geq 1$ ,  $H_r(0) = 0$  and*

$$\begin{aligned} H'_r - \left( \frac{1}{x} + \frac{2x}{1-x} \right) H_r &= x \sum_{j=1}^{r-1} \binom{r}{j} H_j H_{r-j} \\ &\quad + rx \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}. \end{aligned} \quad (3)$$

This triangular system determines every factorial moment. In particular, for  $n \geq 2$ ,

$$\mathbb{E}M_n = \sum_{j=1}^{n-1} (n-j) \frac{(-1)^{j+1} 2^{j-1}}{j!}. \quad (4)$$

If  $a = e^{-2}$  and  $\alpha = (1-a)/2$ , then

$$\mathbb{E}M_n = \alpha n - a + o(1), \quad \text{Var}(M_n) = a^2 n + 2a^2 + o(1). \quad (5)$$

The support is  $\{0\}$  for  $n = 0, 1$ . For  $n \geq 2$ ,

$$\text{supp}(M_n) = \left\{ \left\lceil \frac{n-1}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (6)$$

For every simple cycle  $n \geq 3$ ,

$$G_n(z) = zF_{n-2}(z), \quad \text{supp}(K_n) = \left\{ \left\lceil \frac{n}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (7)$$

Consequently

$$\mathbb{E}K_n = 1 + \mathbb{E}M_{n-2} = \alpha n + o(1), \quad (8)$$

$$\mathbb{E}K_n - \mathbb{E}M_n = e^{-2} + o(1), \quad (9)$$

$$\text{Var}(K_n) = \text{Var}(M_{n-2}) = e^{-4}n + o(1). \quad (10)$$

Both expected occupied-vertex fractions tend to  $1 - e^{-2}$ .

### 3 First-edge decomposition and the Riccati OGF

The first edge in any order is accepted. Conditional on  $e_i$  being first, its endpoints  $i, i+1$  are blocked and the eligible residual vertices split into  $P_{i-1}$  and  $P_{n-i-1}$ . Given this event, the relative orders induced on those two disjoint edge sets are independent and uniform. Their interleaving has no effect because the vertex sets are disjoint. Therefore

$$M_n \stackrel{d}{=} 1 + M_{i-1}^{(L)} + M_{n-i-1}^{(R)} \quad \text{given } e_i \text{ first,}$$

where the two variables on the right are independent. Averaging over the  $n-1$  equally likely choices proves (1). Multiplication by  $x^{n-1}$  and summation over  $n \geq 2$  gives

$$F_x - \frac{F-1}{x} = zxF^2,$$

which is (2). This is a formal-power-series derivation, so it does not presume analytic convergence in two variables.

At  $z = 1$ , normalization gives  $F_n(1) = 1$ , hence  $F(x, 1) = (1-x)^{-1}$ . Differentiate  $zF^2$  exactly  $r$  times:

$$\partial_z^r (zF^2)|_{z=1} = \sum_{j=0}^r \binom{r}{j} H_j H_{r-j} + r \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}.$$

The  $j = 0, r$  terms in the first sum are  $2H_0H_r$ . Moving them to the left proves the all- $r$  equation (3); all other indices are strictly below  $r$ .

For  $r = 1$ , the equation is

$$H_1' - \left( \frac{1}{x} + \frac{2x}{1-x} \right) H_1 = \frac{x}{(1-x)^2},$$

and its formal solution regular at zero is

$$H_1(x) = \frac{x(1 - e^{-2x})}{2(1-x)^2}. \quad (11)$$

Expanding the exponential and extracting  $[x^n]$  proves (4).

## 4 Second factorial moment and variance

For  $r = 2$ , the triangular equation becomes

$$H_2' - \left( \frac{1}{x} + \frac{2x}{1-x} \right) H_2 = 4xH_0H_1 + 2xH_1^2. \quad (12)$$

The unique formal solution regular at zero is

$$H_2(x) = \frac{xe^{-4x}(4x^2e^{2x} - 3xe^{4x} - x + e^{4x} - 1)}{4(x-1)^3}. \quad (13)$$

Direct differentiation verifies (12). The formula also follows by multiplying by the integrating factor  $e^{2x}(1-x)^2/x$  and integrating from zero.

To extract the variance without fitting finite data, set  $s = 1 - x$ ,  $a = e^{-2}$ , and  $\alpha = (1 - a)/2$ . Equations (11) and (13) have pole parts

$$H_1 = \frac{\alpha}{s^2} - \frac{1+a}{2s} + R_1(x), \quad (14)$$

$$H_2 = \frac{c_3}{s^3} + \frac{c_2}{s^2} + \frac{c_1}{s} + R_2(x), \quad (15)$$

where

$$c_3 = \frac{(1-a)^2}{2}, \quad c_2 = -\frac{5}{4} + a + \frac{5}{4}a^2, \quad c_1 = \frac{3}{4} + a + \frac{5}{4}a^2. \quad (16)$$

After these pole parts are removed,  $R_1, R_2$  are entire combinations of exponentials and polynomials divided by removable powers of  $1 - x$ . Their Taylor coefficients are therefore  $o(1)$  (indeed factorially small up to a polynomial factor). It follows that

$$\begin{aligned} \mathbb{E}M_n &= \alpha(n+1) - \frac{1+a}{2} + o(1) = \alpha n - a + o(1), \\ \mathbb{E}(M_n)_2 &= c_3 \binom{n+2}{2} + c_2(n+1) + c_1 + o(1). \end{aligned}$$

Since  $c_3/2 = \alpha^2$ , substituting these expressions into  $\text{Var}(M_n) = \mathbb{E}(M_n)_2 + \mathbb{E}M_n - (\mathbb{E}M_n)^2$  cancels the quadratic term. The linear coefficient simplifies to  $a^2$  and the constant to  $2a^2$ , proving (5). Thus the requested leading law is  $\text{Var}(M_n) = e^{-4}n + O(1)$ , with its boundary constant also identified.

## 5 Exact support: binary gap words

**Lemma 1** (Path support). *A cardinality  $k$  occurs as a maximal matching of  $P_n$  exactly when  $2k \leq n \leq 3k + 1$ , with the separate empty cases  $n = 0, 1$ .*

*Proof.* List the  $k$  selected dimers from left to right. There are  $k + 1$  gaps of unmatched vertices: two end gaps and  $k - 1$  internal gaps. Maximality is equivalent to every gap having size at most one, since two adjacent unmatched vertices would support another edge. Hence

$$n = 2k + g_0 + \cdots + g_k, \quad g_j \in \{0, 1\},$$

which is possible exactly for  $0 \leq n - 2k \leq k + 1$ . Conversely, choosing any  $n - 2k$  of these gaps to have size one explicitly constructs a maximal matching. This proves the equivalence.  $\square$

Every such maximal matching occurs with positive probability under RSA: place all of its mutually disjoint edges first in the scan. They are accepted, and every remaining edge meets one of them by maximality, so it is rejected. Lemma 1 therefore proves (6), not merely a bound on possible sizes.

For a cycle with  $k$  selected dimers there are  $k$  cyclic gaps, each again zero or one. Thus  $n = 2k + g_1 + \cdots + g_k$ , so  $2k \leq n \leq 3k$ , and the same ordering argument realizes every constructed matching. This proves the support in (7). Notice that smaller support values are maximal but not maximum; for example, the middle edge alone is a maximal matching of  $P_4$ , while a maximum matching has size two.

## 6 Cycle identity and the boundary correction

The first edge of  $C_n$  is accepted. Removing its two endpoints from further competition leaves exactly a path on  $n - 2$  vertices; conditional on the first edge, the residual relative order is uniform. The identity  $K_n \stackrel{d}{=} 1 + M_{n-2}$  follows for every  $n \geq 3$ , including  $C_3$  and  $C_4$ , and proves (7). Taking moments gives (8) and  $\text{Var}(K_n) = \text{Var}(M_{n-2})$ . Finally,

$$1 + \alpha(n - 2) - a = \alpha n, \quad 1 - 2\alpha - a = 0,$$

which explains the exact cancellation of the path mean’s constant boundary term. Comparison with the same-size path leaves the positive correction  $a = e^{-2}$  in (9). The cycle variance has no constant term at this precision because  $a^2(n - 2) + 2a^2 = a^2n$ .

## 7 Boundary atlas and theorem limits

| Face           | Exact treatment          | Not silently identified with   |
|----------------|--------------------------|--------------------------------|
| $P_0, P_1$     | $M_n = 0, F_n = 1$       | the $n \geq 2$ recurrence      |
| $P_2$          | $M_2 = 1, F_2 = z$       | a limiting-density statement   |
| $C_n$          | simple cycle, $n \geq 3$ | loops or parallel-edge cycles  |
| priority ties  | null under continuity    | unspecified discrete tie rules |
| terminal state | maximal matching         | guaranteed maximum matching    |
| finite tables  | regression evidence      | proof for arbitrary $n$        |

The theorem concerns the final count. It does not classify adsorption times under an additional physical-time embedding, general graphs, weighted edge orders, correlated priorities, or discrete ties. Such changes destroy at least one symmetry used in the first-edge convolution.

## 8 Executable reconstruction and adversarial controls

The evidence producer implements (1) in exact rational arithmetic through  $P_{20}$ , stores orders zero through five of the factorial triangle, and independently propagates the first two moments through  $n = 200$ . It records full path distributions through  $P_{10}$ , cycle distributions through  $C_9$ , and asymptotic cells at  $n = 20, 50, 100, 200$ .

The checker imports no producer module and never trusts a floating Monte Carlo sample. Its state is a pair of bitmasks: the set of processed labeled edges and the set of matched vertices. Extending every state by every unprocessed edge aggregates all 818,225 labeled orders in the declared window and recovers their exact integer multiplicities. A separate implementation of the falling-factorial conditional identity reconstructs every stored high-order moment. In the frozen release it executes 19,371 assertions.

A SymPy lane checks the Riccati specialization, the  $H_1, H_2$  differential equations, pole algebra, finite coefficient controls, and path-to-cycle support shift in 132 symbolic checks. Two unrelated temporary paths reproduce the evidence byte for byte. Strict JSON and recursive safe-YAML loaders reject duplicate keys, nonstandard constants, unknown or missing keys, bool-as-integer confusion, row-order or uniqueness loss, and any semantic drift; exact top/nested YAML schemas, types, values, and a semantic hash are release-blocking. All 105/105 repaired-hash, drop-replace, stale-hash, raw-duplicate, nonstandard-constant, and YAML attacks are rejected.

These finite calculations are falsification tools. The all- $n$  claims rest on the first-edge decomposition, formal differentiation, pole extraction, and binary-gap proof above. No finite cutoff is promoted to an induction proof.

## 9 Route-A assessment and nonclaims

| Gate | Verdict | Strongest obstruction                                       |
|------|---------|-------------------------------------------------------------|
| A0   | FAIL    | no rational-prime carrier or arithmetic weight              |
| A1   | FAIL    | terminating random scans are not primitive periodic orbits  |
| A2   | FAIL    | a PGF is source probability data, not a target determinant  |
| A3   | FAIL    | no divisor, functional equation, or explicit-formula bridge |
| A4   | FAIL    | no source-native same-clock self-adjoint quantization       |

The tuple is (A0\_FAIL, A1\_FAIL, A2\_FAIL, A3\_FAIL, A4\_FAIL), so the overall verdict is ROUTE\_A\_REJECTED. Route B is not authorized. In particular, we do not reinterpret random histories as an unweighted orbit product, attach

prime labels to graph edges, call a PGF an Euler factor, or introduce a formal quantum operator. The governing obstruction is HEN-O275, within the literal scope NO\_BAD\_EULER\_OR\_ROOT\_NUMBER.

## 10 Data, code, and research declarations

**Data and code availability.** All data are generated deterministically in the package. The canonical evidence JSON, producer, independent checker, SymPy reconstruction, fresh-path replay, hostile mutation suite, manuscript source, three revision PDFs, and self-excluded release ledger are included. No external dataset or training data are used.

**Ethics.** This is a theoretical and computational study of a finite stochastic model. It uses no human participants, animals, personal data, or clinical decisions; institutional ethics approval is not applicable.

**Conflicts of interest.** No conflicts of interest are declared.

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**CRedit authorship.** The package records one accountable computational authorship role covering conceptualization, formal analysis, software, validation, visualization audit, and writing. Classical ownership is assigned through the cited literature.

**AI-use disclosure.** AI assistance was used to organize exposition and executable checks. Every mathematical claim is supported by the displayed derivation and deterministic reconstruction; responsibility for verification remains with the package author. AI assistance is not listed as an author and no claim of autonomous literature originality is made.

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